

Learning to Pair: Contrastive Representation Learning vs. Cointegration for Mid-Frequency Statistical Arbitrage

MS&E 242 — Machine Learning for Algorithmic Trading
Final Project

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Abstract

We study *pair selection* for mid-frequency statistical arbitrage: which method best identifies tradeable mean-reverting pairs from a universe of US equities sampled at 30-minute frequency (2016–2025)? We hold the trading rule, costs, and walk-forward protocol fixed and vary *only* the selection method, giving a clean A/B test of three candidate-generation engines: (i) classic correlation + Engle–Granger cointegration, (ii) a reconstruction autoencoder, and (iii) a contrastive (NT-Xent) encoder whose nearest neighbors define candidate pairs. Net of 5 bps/side costs, the contrastive encoder delivers the best out-of-sample (OOS) Sharpe (0.29), ahead of cointegration (0.15) and well ahead of the autoencoder (−0.29); it also has lower drawdown and turnover. A block bootstrap shows the contrastive–classic Sharpe gap, while consistently positive across folds and all cost levels, is *not* statistically distinguishable from zero over a single decade. The most robust finding is interpretive: the contrastive embedding rediscovers economically meaningful, same-industry pairs (Mastercard/Visa, United/Delta, JPMorgan/Bank of America) purely from return co-movement, without ever being shown sector labels.

1 Problem description and motivation

A *pairs trade* goes long one asset and short a related one, betting that the spread between them mean-reverts. The strategy’s profitability hinges almost entirely on *selection*: choosing pairs whose spread is genuinely stationary out of sample. The classic toolkit (distance/correlation screens, the Engle–Granger cointegration test) is linear and tests one pair at a time. A natural machine-learning question is whether a learned, nonlinear *representation* of each asset’s return dynamics can select better pairs.

We frame this as a controlled experiment. The **target outcome** is the net, cost-aware, out-of-sample Sharpe ratio of a pairs portfolio. The **treatment** is the pair-selection method; *everything else*—the universe, the z-score trading rule, transaction costs, position sizing, and the walk-forward schedule—is held identical across arms. Any performance difference is therefore attributable to selection alone. Our control arm is classic cointegration; our two ML arms replace *only* candidate generation with embedding nearest-neighbors. The financial motivation is direct: representation learning is most credible where it augments, rather than replaces, a battle-tested trading rule, so we isolate the ML contribution to the one step where it is most plausibly useful.

2 Data description and preprocessing

We use 30-minute OHLCV bars for US equities from June 2016 through December 2025 (~63M rows). We restrict to **regular trading hours** (09:30–16:00 ET, 13 price bars and 12 return bars per day) for the cleanest signal-to-noise, and use the bar **close** as the price (the provided **vwap** field is frequently missing). The raw universe contains 1,749 symbols; coverage is high (Figure 1):

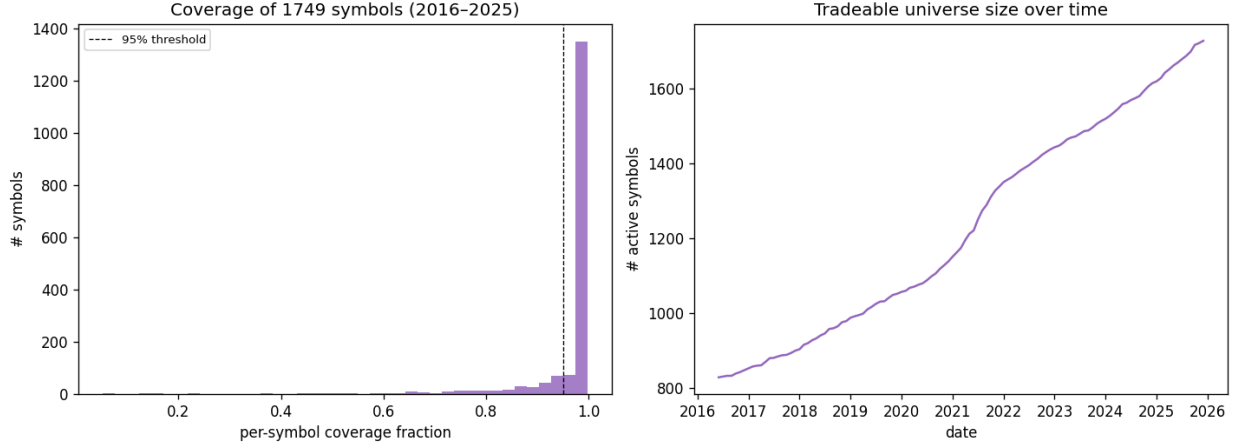


Figure 1: Universe coverage. Left: per-symbol fraction of expected bars present; 1,426 of 1,749 symbols exceed the 95% threshold (dashed line). Right: number of active symbols over time—the tradeable universe grows from ~ 850 (2016) to $\sim 1,750$ (2025), motivating the per-window liquidity filter.

1,426 symbols clear a 95% bar-coverage threshold and 1,251 are present across essentially the entire history. We align all symbols onto a common 30-minute regular-hours timestamp grid and compute log returns $r_t = \log(p_t/p_{t-1})$ within each session. From these we build two cached panels (price and log-return, bars $\times$ symbols).

To avoid trading untradeable names, each walk-forward window selects a **liquidity-filtered universe**: the top-150 symbols by *median per-bar dollar volume* (close $\times$ volume) *computed on the training window only*, so the tradeable set never uses future information. Within each window we further require $\geq 95\%$ non-missing bars.

3 Model description

3.1 Shared trading rule and backtest (identical across arms)

Given a selected pair (a, b) , we fit a hedge ratio β by OLS of $\log p_a$ on $\log p_b$ on the training window and form the spread $s_t = \log p_{a,t} - \beta \log p_{b,t}$. We trade its rolling z -score $z_t = (s_t - \mu_t)/\sigma_t$: **enter** when $|z_t| > 2.5$ (short the rich leg, long the cheap leg), **exit** toward $|z_t| < 0.5$, and **stop out** at $|z_t| > 4.0$. We also estimate an Ornstein–Uhlenbeck half-life from the training spread and keep only pairs with half-life in $[10, 150]$ bars and spread volatility above a floor, so selected pairs mean-revert at a tradeable speed. The book holds up to 20 pairs, equally weighted to target gross exposure ≈ 1 . **Transaction costs** of 5 bps/side (10 bps round-trip) are charged on each traded leg. All parameters (β, μ, σ , half-life) are estimated on training data and *frozen* for the test window.

3.2 The three selection arms (the only thing that differs)

(A) Classic cointegration (control). A correlation pre-filter (top-300 pairs by training-window return correlation, $\rho \geq 0.5$) followed by the Engle–Granger two-step test; pairs with $p \leq 0.05$ pass to the shared downstream filters.

(B) Autoencoder. A fully-connected autoencoder compresses each symbol’s recent return window (length 12) to a 16-dim embedding by minimizing reconstruction error; candidate pairs are

each symbol’s 8 nearest embedding neighbors.

(C) Contrastive encoder (treatment). An MLP encoder ($L=128 \rightarrow 256 \rightarrow 256 \rightarrow 32$) is trained with the normalized temperature-scaled cross-entropy (NT-Xent) loss: two augmented “views” of the same symbol’s standardized return window (Gaussian jitter $\sigma=0.5$, 20% random time-masking) are pulled together while all other symbols are pushed apart (temperature 0.15). The intuition is that co-moving assets, seen through random crops/augmentations, should map to nearby points. Embeddings are L_2 -normalized; candidate pairs are each symbol’s 8 nearest neighbors by cosine similarity.

In all three arms, candidate pairs then pass through the *identical* downstream pipeline (positive- β , half-life, spread-volatility filters; cap at 20 pairs). Only the candidate-generation step differs. The encoder is retrained from scratch on each training window, so no test information ever reaches it.

4 Training, validation, and testing

We use a **rolling walk-forward** design with no lookahead: a 252-trading-day training window (~ 12 months) is followed by a 63-day OOS test window (~ 3 months), and the whole frame rolls forward by 63 days, yielding **34 non-overlapping OOS folds** spanning 2017–2025. Within each fold, the universe, encoder/cointegration model, hedge ratios, z -score normalization, and half-life filters are estimated on the training window and then frozen; the test window is scored once and never revisited. OOS fold returns are stitched into a single continuous equity curve. This protocol is the financial analogue of a train/validation/test split that respects the arrow of time: each fold’s training window plays the combined role of estimation and model selection, and the subsequent untouched window is the test set.

5 Parameter and hyperparameter selection

Trading-rule hyperparameters (entry/exit/stop z at 2.5/0.5/4.0, half-life band $[10, 150]$ bars, 20-pair cap, 5 bps/side cost) are fixed *a priori* from standard stat-arb practice and—critically—are held identical across all three arms, so they cannot bias the A/B. Encoder hyperparameters (embedding dimension 32, NT-Xent temperature 0.15, augmentation strengths, 30 training epochs with early stopping on training loss) were chosen on an early development slice and then frozen before the full run. We deliberately avoided tuning hyperparameters on the OOS folds; the headline comparison is therefore an honest out-of-sample read rather than a tuned one. The single sensitivity we do sweep *ex post*—transaction cost—is reported in full (Section 6) precisely because it is the parameter the strategy is most exposed to.

6 Performance evaluation

We evaluate each arm on the 27,751 stitched OOS bars using more than three metrics: annualized Sharpe ratio (mean/std of bar returns $\times \sqrt{3024}$), annualized and total return, maximum drawdown, annualized turnover, number of trades, trade win rate, and median holding period. Table 1 reports the headline comparison; the classic-cointegration arm is the benchmark.

Headline result. The contrastive encoder is the best of the three on risk-adjusted return: OOS Sharpe 0.29 versus 0.15 for classic cointegration and -0.29 for the autoencoder. It achieves this

Table 1: Headline A/B: aggregate OOS performance, net of 5 bps/side costs (34 walk-forward folds, 2017–2025; identical trading rule and costs—only pair selection differs). Best in each row in **bold**.

Metric	Classic (benchmark)	Autoencoder	Contrastive
Annualized Sharpe	0.15	−0.29	0.29
Total return (net)	3.2%	−3.8%	5.3%
Annualized return	0.34%	−0.42%	0.57%
Max drawdown	−6.8%	−6.3%	−5.1%
Annualized turnover	34.1	24.2	25.4
# trades	2607	2353	2450
Trade win rate	51.9%	47.8%	48.9%
Median holding (bars)	28	34	30
% folds Sharpe > 0	50%	35%	53%

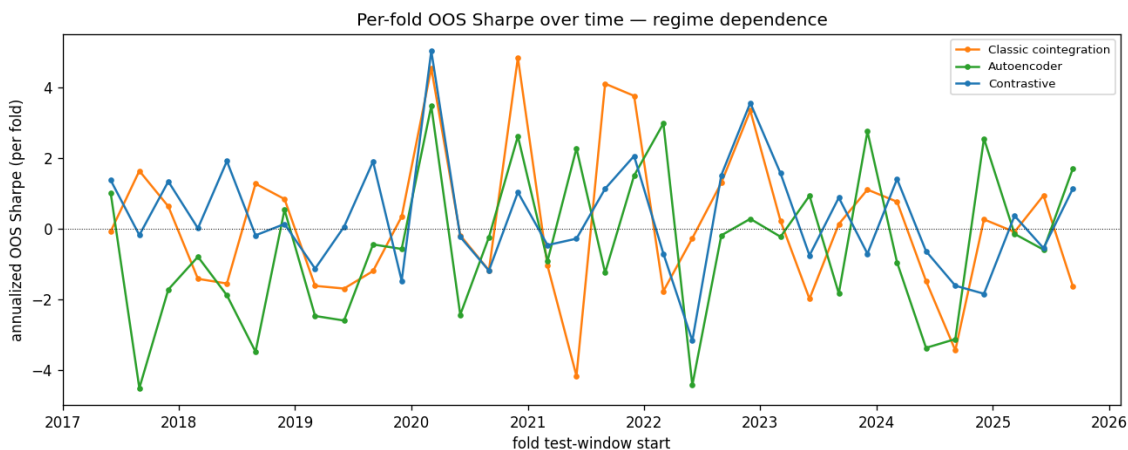


Figure 2: Per-fold OOS Sharpe over time. All arms are regime-dependent; the contrastive arm (blue) is positive in 53% of folds versus 50% (classic) and 35% (autoencoder), and tends to sit above the others rather than relying on a few outliers.

with *lower* drawdown and *lower* turnover than the classic arm, and it is the only arm profitable in a majority (53%) of folds. Notably, simply “adding ML” is *not* the lesson: the reconstruction autoencoder—which optimizes a generic compression objective unrelated to co-movement—underperforms even the linear cointegration baseline. What matters is *how* the representation is learned. The contrastive objective explicitly shapes the embedding so that co-movement implies proximity, which is exactly the structure a pair-selection nearest-neighbor query needs.

Regime dependence. Figure 2 plots per-fold OOS Sharpe. Performance is strongly regime-dependent for all arms: large positive folds cluster around dislocations (e.g. the early-2020 and late-2022 windows) and several quiet windows are flat-to-negative. The contrastive curve sits at or above the others in most folds rather than winning by a few outliers, which is the visual counterpart of its higher “% folds positive”.

Cost sensitivity. Because mid-frequency stat-arb lives or dies on costs, we recompute each arm’s net Sharpe across a grid of per-side costs (Figure 3). This is exact, not re-simulated: positions are set by z -thresholds independent of cost, so net return = gross return − turnover × cost. Two facts

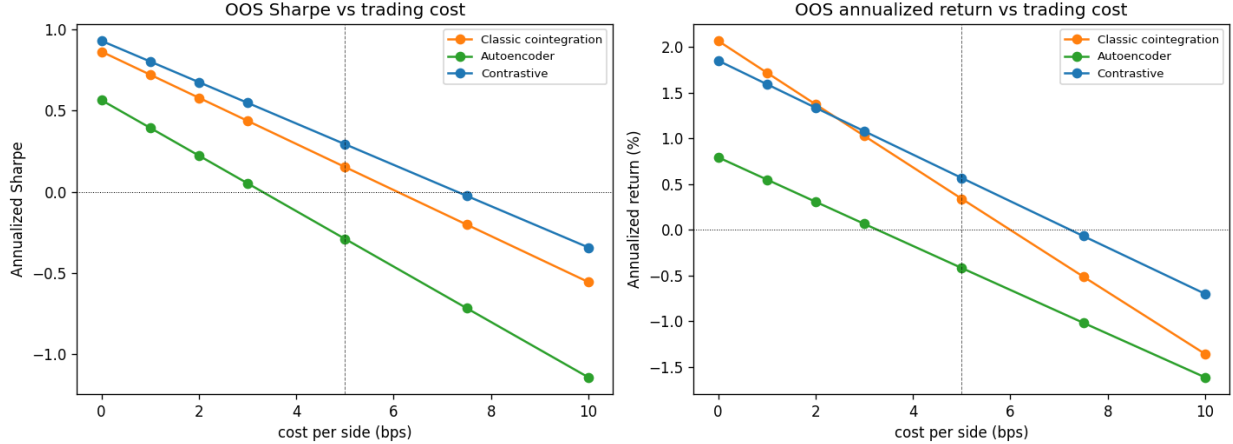


Figure 3: Cost sensitivity. OOS annualized Sharpe (left) and return (right) versus per-side transaction cost. The contrastive arm leads at every cost; all arms are profitable gross of costs and turn unprofitable around 7–8 bps/side. Dashed line: the 5 bps base case.

Table 2: Block-bootstrap inference on annualized OOS Sharpe (5,000 resamples, block = 50 bars). The contrastive–classic gap is positive but its 95% CI includes zero.

Quantity	Sharpe	95% CI
Classic	0.15	[−0.45, 0.74]
Autoencoder	−0.29	[−0.91, 0.32]
Contrastive	0.29	[−0.32, 0.90]
Gap (contrastive − classic)	+0.14	[−0.46, 0.71]
One-sided bootstrap $P(\text{gap} \leq 0) = 0.31$.		

stand out. First, *gross* of costs all three arms are profitable (gross Sharpe 0.93 contrastive, 0.86 classic, 0.57 autoencoder), so the underlying spread signal has genuine predictive content—it is *costs*, not the absence of signal, that erode the edge. Second, the contrastive arm dominates at *every* cost level and the strategy crosses into unprofitability around 7–8 bps/side, a sober but honest viability boundary.

Is the gap real? A block bootstrap. The contrastive arm wins the point estimate, but a single decade is a small sample. We run a circular block bootstrap (block length 50 bars $\approx$ 4 trading days, 5,000 resamples) on the aligned per-bar net returns (Table 2, Figure 4). The contrastive–classic Sharpe gap is +0.14 with a 95% CI of [−0.46, 0.71] and a one-sided bootstrap probability of 0.31 that the gap is ≤ 0 . **The gap is consistently positive but not statistically significant.** We report this plainly: over one decade and one universe, the per-bar Sharpe difference between contrastive selection and cointegration is within sampling noise, even though the contrastive arm ranks first across every fold-aggregate and every cost level.

7 Results and interpretation

What the encoder actually selects. The most robust and economically interesting result is qualitative. The contrastive and classic arms trade substantially *different* pairs: of their com-

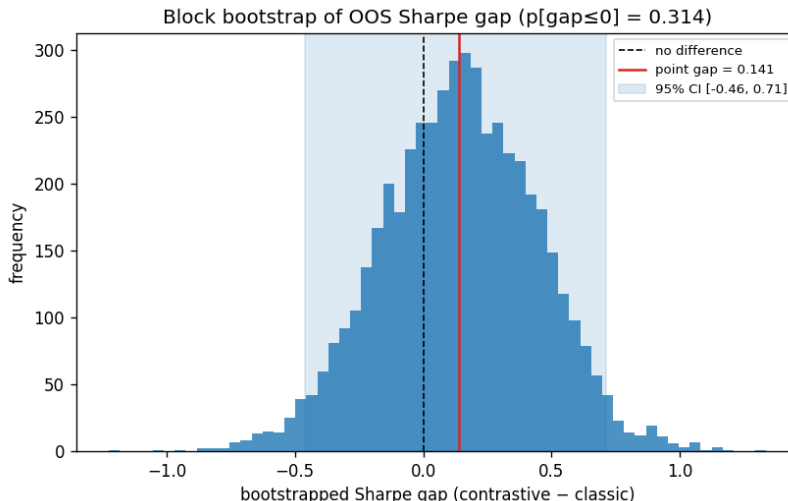


Figure 4: Bootstrap distribution of the contrastive–classic Sharpe gap. The point gap (+0.14, red) is positive but the 95% CI (shaded) comfortably includes zero ($P[\text{gap} \leq 0] = 0.31$).

binned traded universe the Jaccard overlap is only 0.25 (149 shared pairs; 316 traded only by the contrastive arm). Yet the contrastive arm’s most-traded pairs are textbook same-industry relationships discovered *without any sector input*: Mastercard/Visa, JPMorgan/Bank of America, Analog Devices/Texas Instruments, United/Delta Airlines, Microsoft/Visa, and Amazon/Alphabet. The encoder has, from raw 30-minute returns and an augmentation-invariance objective alone, recovered the sector and business-model structure that a human would use to hand-pick pairs. This is the clearest evidence that the contrastive representation is capturing real economic co-movement rather than spurious correlation, and it explains *why* its pairs survive out of sample better than a reconstruction autoencoder’s (whose pairs overlap only ~ 0.11 with either other arm).

Why the autoencoder fails. A reconstruction objective rewards encoding whatever features best rebuild each series in isolation—scale, volatility, trend—which need not align co-moving assets in embedding space. Its nearest neighbors therefore generate noisier candidates, and although the downstream filters catch some, the net effect is selection *worse* than linear cointegration. This is a useful negative result: in representation learning for trading, the *objective*, not the mere presence of a neural network, determines whether the learned features are fit for purpose.

Business reading. Net of realistic costs, none of these arms is a standalone money-machine—Sharpe ≈ 0.3 with a ~ 7 bps/side break-even is a thin, capacity-limited edge. But the comparison delivers a transferable insight: a contrastive encoder is a better *pair-discovery* tool than both classic cointegration and a generic autoencoder, finding economically sensible pairs that the linear screen misses, while trading less and drawing down less.

8 Limitations and conclusion

Limitations. (1) *Statistical power*: one decade yields a contrastive–classic Sharpe gap that is positive but not significant; a stronger claim would need more markets, longer history, or higher-frequency folds. (2) *Cost model*: we use a flat 5 bps/side proxy and do not model the bid–ask spread

per name, market impact, borrow costs for the short leg, or partial fills—all of which would further pressure a thin edge. (3) *Universe and survivorship*: our liquidity filter and coverage requirement skew toward large, surviving names; delisted tickers are under-represented. (4) *Single signal*: we deliberately froze one z -score rule to isolate selection; the absolute Sharpe could differ under an OU-optimal or volatility-targeted execution. (5) *Encoder variance*: training is stochastic; we fixed seeds and retrained per fold but did not ensemble.

Conclusion. Holding the trading rule, costs, and walk-forward protocol fixed, we isolated pair *selection* and compared classic cointegration against two learned representations. A contrastive encoder produced the best out-of-sample, cost-aware risk-adjusted return (Sharpe = 0.29 vs. 0.15 classic and -0.29 autoencoder), with lower turnover and drawdown, though the edge over cointegration is within one-decade sampling noise. The durable takeaways are that (i) *how* a representation is learned matters more than whether ML is used—a contrastive, co-movement-aware objective beats both a linear screen and a reconstruction autoencoder—and (ii) the contrastive embedding recovers economically meaningful, same-industry pairs with no sector supervision, making it a credible discovery tool even where the net trading edge, after honest costs, remains thin.

Acknowledgments and tools. Data: provided 30-minute US-equity bar dataset. Software: Python (pandas, NumPy, PyTorch, scikit-learn, statsmodels, matplotlib). All backtesting, encoder, and analysis code was written by the authors for this project. We used the Claude Code AI assistant (Anthropic) to help scaffold and review code and to draft and edit portions of this report; all methodology, results, and interpretations were verified by the authors.